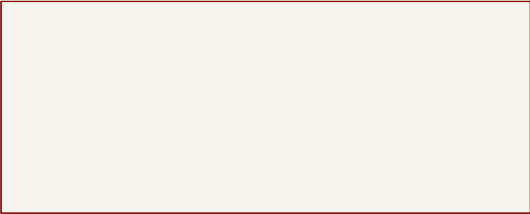


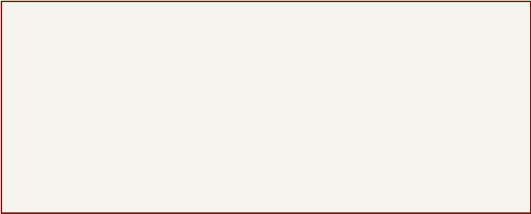
CONTROLLER – SEVEN READABLE KICAD SHEETS

01 Power and Pi



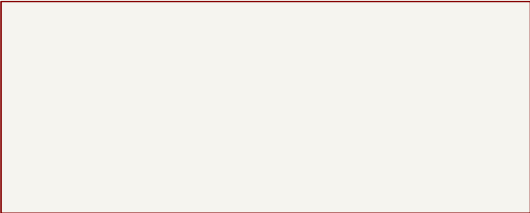
File: 01_power_and_pi.kicad_sch

02 ESP32 and Pi UART



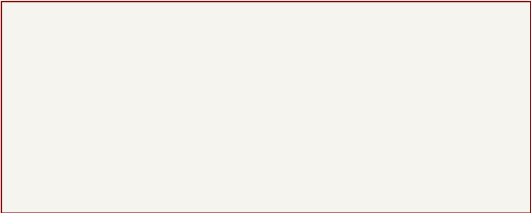
File: 02_esp32_and_pi.kicad_sch

03 Left Motor



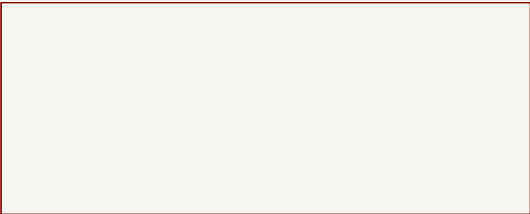
File: 03_left_motor.kicad_sch

04 Right Motor



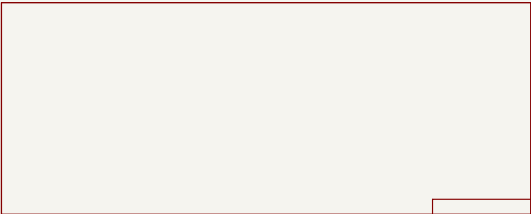
File: 04_right_motor.kicad_sch

05 Encoders



File: 05_encoder_inputs.kicad_sch

06 Arm Servos



File: 06_arm_servos.kicad_sch

07 Main Servo 8.4V



Harrison

Sheet: /
File: pickleball_robot_controller.kicad_sch

Title: Pickleball Robot Controller – Hierarchical Overview

Size: A4	Date: 2026-07-12	Rev: R2-MULTISHEET
KiCad E.D.A. 10.0.4		Id: 1/8

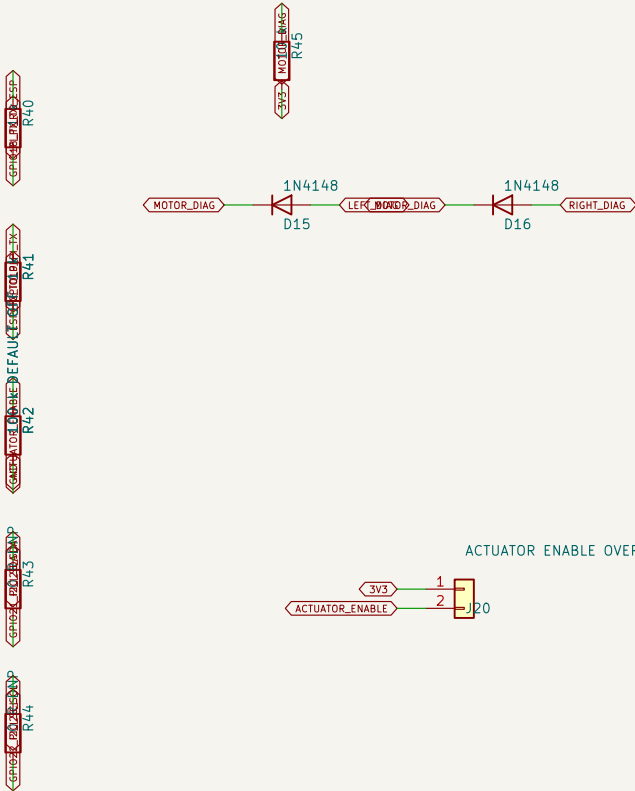
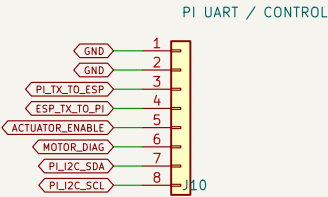
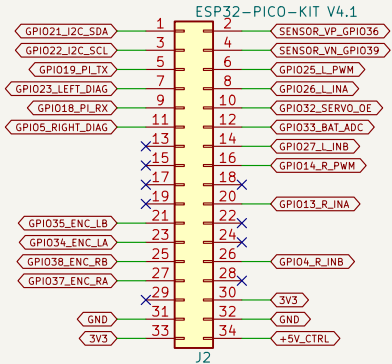
BATTERY INPUT AND PROTECTION

RASPBERRY PI 5 V / 5 A BUCK

J18 must be closed by a latching E-stop switch. It removes actuator power but keeps Pi/ESP logic alive.

BATTERY VOLTAGE MONITOR

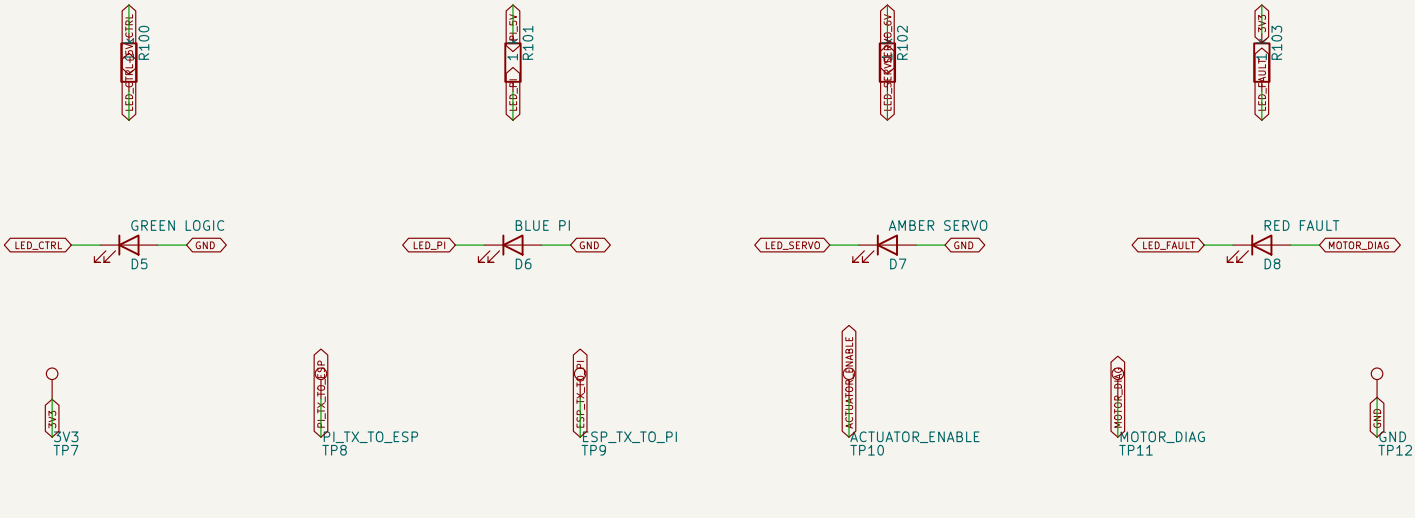
5 V CONTROL BUCK – ESP32 AND LOGIC ONLY



J10: 1/2 GND; 3 Pi TX->ESP GPIO18; 4 ESP GPIO19->Pi RX; 5 actuator enable; 6 fault; 7/8 I2C DNP.

Pi: GPIO14 pin 8 -> J10.3; GPIO15 pin 10 <- J10.4; GPIO17 pin 11 -> J10.5; GND -> J10.1/2.

STATUS LEDS



LEFT WHEEL MOTOR DRIVER

5 A MINI BLADE MOTOR BRANCH
P5

LEFT_INB_DRV
R80

LEFT_INB_DRV
R81

LEFT_INB_DRV
R82

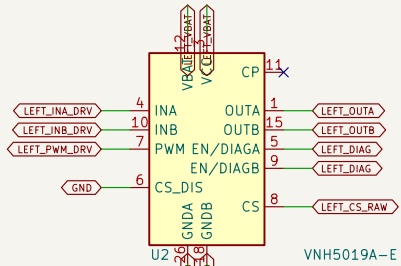
3V3
LEFT_PWM_DRV
R86

ENC_A_CABLE
R83

ENC_A_CABLE
R84

ENC_A_CABLE
R85

3V3
ENC_B_CABLE
R89

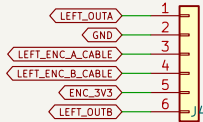


470 uF 35 V LOW ESR
C12

100 nF 50 V
C13

1 uF 25 V
C14

LEFT MOTOR + ENCODER



LEFT MOTOR LARGE SOLDER PADS



100 nF 50 V FIT AT MOTOR
C18



10 nF current filter
C80



J4/J5 cable: 1 RED motor+; 2 BLACK encoder GND; 3 YELLOW encoder A; 4 GREEN encoder B; 5 BLUE encoder +3V3; 6 WHITE motor-.

J15: large 1.7 mm through-holes in parallel with pins 1/6 for separately soldered red and white motor-power wires.

VBAT
TP20

PWM
TP21

INA
TP22

INB
TP23

DIAG
TP24

CS_ADC
TP25

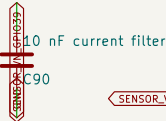
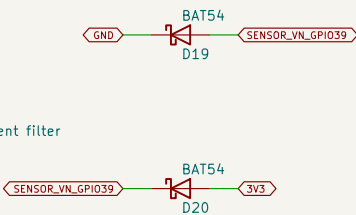
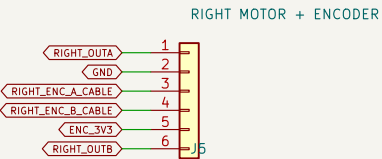
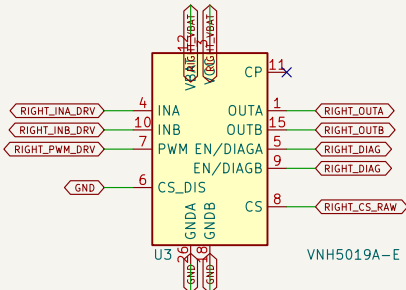
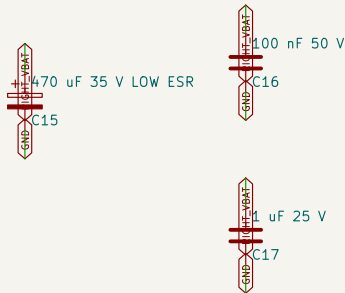
ENC_A
TP26

ENC_B
TP27

GND
TP28

RIGHT WHEEL MOTOR DRIVER

5 A MINI BLADE MOTOR BRANCH
P6



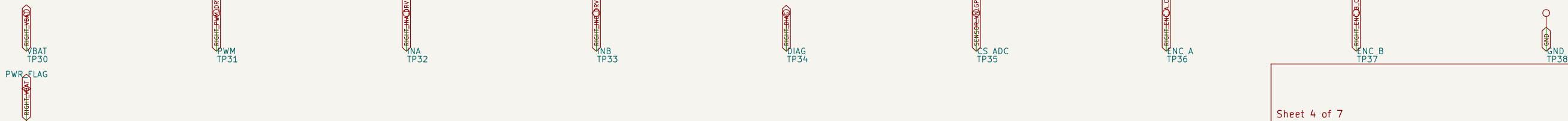
C protect current sense R14

3V3 pullup R96

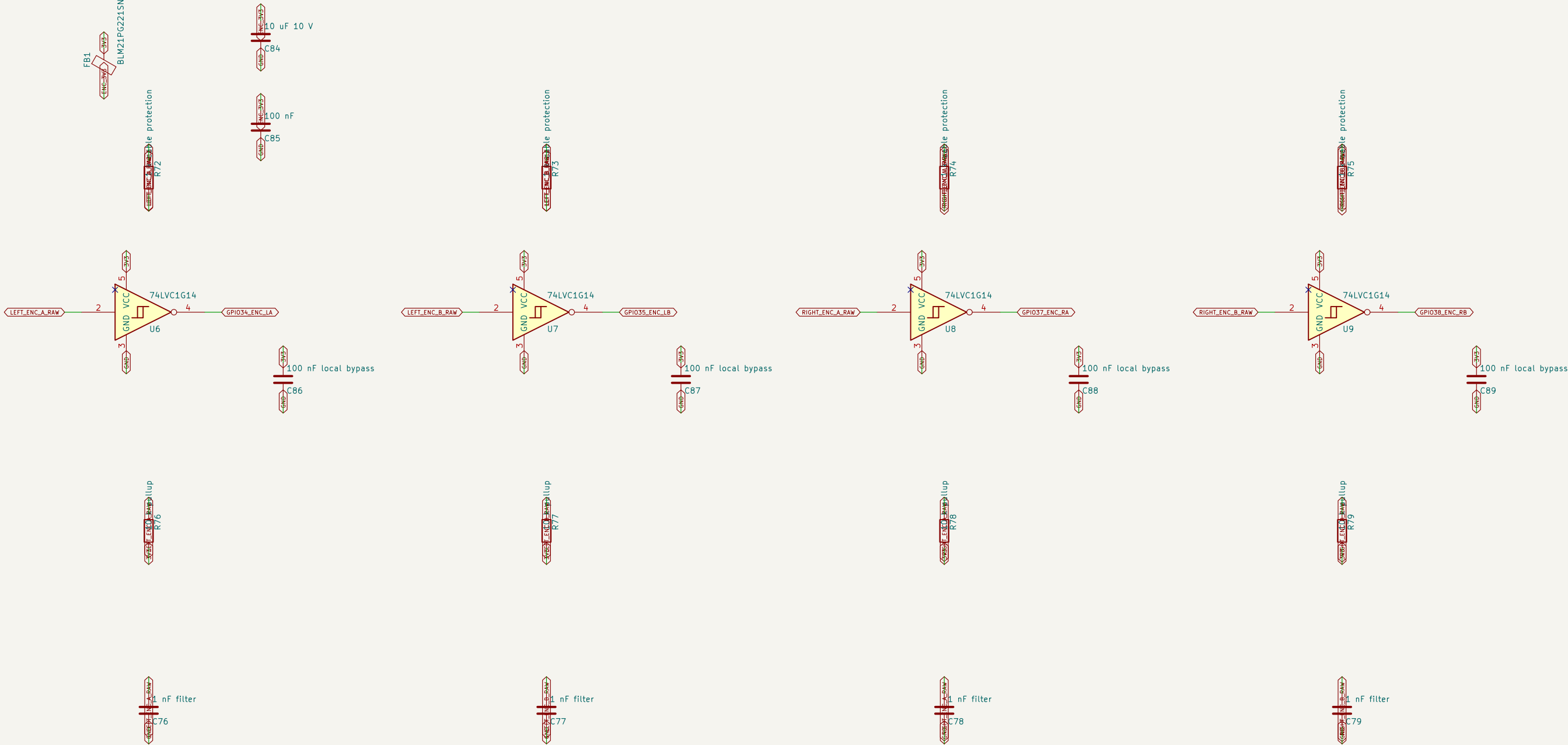
series R99

J4/J5 cable: 1 RED motor+; 2 BLACK encoder GND; 3 YELLOW encoder A; 4 GREEN encoder B; 5 BLUE encoder +3V3; 6 WHITE motor-.

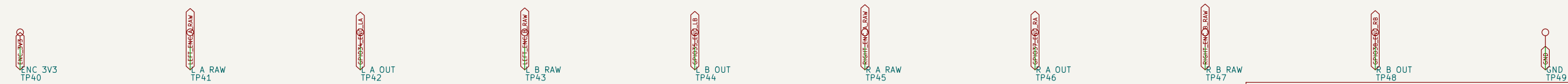
J16: large 1.7 mm through-holes in parallel with pins 1/6 for separately soldered red and white motor-power wires.



FOUR 3.3 V SCHMITT-BUFFERED ENCODER INPUTS



The six-pin J4/J5 connectors are on the motor-driver sheets; their encoder A/B pins enter here.



Sheet 6 of 7		
Harrison		
Sheet: /06 Arm Servos/ File: 06_arm_servos.kicad_sch		
Title: Servo Power Regulator and 3-DOF Arm Control		
Size: A3	Date: 2026-07-12	Rev: R2-MULTISHEET
KiCad E.D.A. 10.0.4		Id: 7/8

DEDICATED 8.4 V / 5 A BUCK FOR MIUZEI 25KG 270 DEG SERVO

